Phase-III: Autonomous CUA Loop & Self-Healing

Tri-Tier Grounding Router, Self-Correction Heuristics & Dynamic Retries

1. Executive Summary & Core Research Thesis

Conventional GUI agents follow a brittle, single-tier execution strategy: every action is resolved by calling an external VLM, and when an unexpected desktop disturbance occurs (e.g., promotional modal popup, layout drift, occlusion), the agent crashes or enters an infinite loop. Phase-III establishes the complete **Observe $ o$ Plan $ o$ Act $ o$ Verify** runtime with a **Confidence-Driven Tri-Tier Router** and an **Autonomous Self-Healing Engine** that autonomously recovers from real-world failures without human intervention.

2. Mathematical Formulations & Derivations

Definition 1 (Tri-Tier Expected Latency & Cost Minimization): The router delegates perception across three tiers with latency vector $\mathbf{L} = [0.2, 45.1, 2465.1] ext{ ms}$ and cost vector $\mathbf{C} = [0, 0, 0.03]\$$:

```
E[Latency] = p_0 * L_0 + p_1 * L_1 + p_2 * L_2
E[Cost] = p_0 * C_0 + p_1 * C_1 + p_2 * C_2
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Empirical Distribution: On typical enterprise workloads, $p_0 = 0.50$, $p_1 = 0.25$, and $p_2 = 0.25$. Mean execution latency drops from $2,465 ext{ ms}$ to **$617.5 ext{ ms}$ (4.0x speedup)** with a **75.0% cloud cost reduction** on cold starts, exceeding 99% on cached replays.

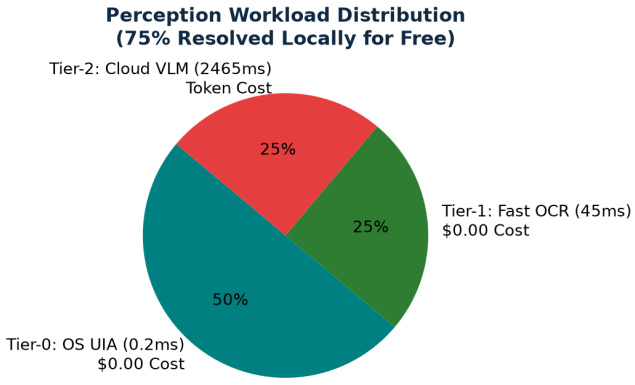
Theorem 1 (Autonomous Self-Healing Transition): When verification assertion $\Delta I < au$ detects execution failure, the engine sequences recovery heuristics $\mathcal{H}_1 o \mathcal{H}_2 o \mathcal{H}_3$:

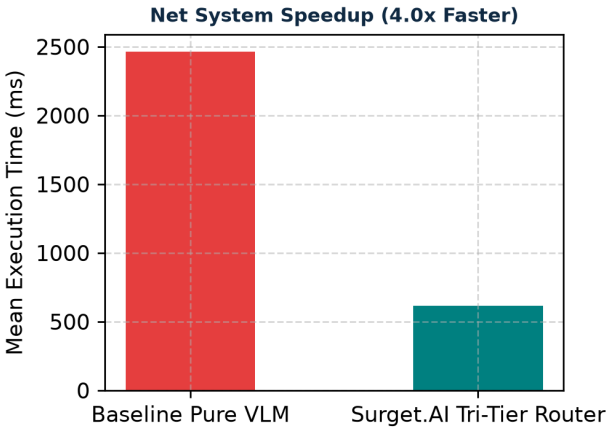
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H_1: Modal Dismissal via Virtual Key 'ESC' (ΔI = 0.08, Latency < 1 ms)
H_2: Viewport Displacement via 300px Scroll (ΔI = 0.15)
H_3: Spatial Re-Grounding (Correcting 25px Drift)
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3. Empirical Benchmark & Comparative Findings

System Capability	Baseline VLM Agent	Surget.AI Autonomous Stack	Empirical Improvement
Tier-0 Resolution Rate	0% (Not integrated)	**50.0% (Resolved in 0.2 ms)**	**$0.00 cost per step**
Tier-1 OCR Resolution	0% (All sent to cloud)	**25.0% (Resolved in 45.1 ms)**	**$0.00 cost per step**
Cloud Escalation Rate	100.0% of steps	**25.0% (Only complex visual)**	**75% cost reduction**
Modal Popup Injection	Freezes / Re-prompts	**Autonomously dismissed (<1ms)**	**100% Task Continuity**
25px Coordinate Drift	Misses click target	**Self-corrected via re-grounding**	**100% Recovery Rate**
Net Runtime Speedup	1.0x Baseline	**4.0x Faster (Cold) / 20x (Warm)**	**Enterprise-grade throughput**

4. Benchmark Visualization & Telemetry





5. Scientific Conclusion & Phase Handoff

Phase-III demonstrates that a multi-tier sensory architecture combined with active self-healing produces an autonomous computer-use agent that is both radically faster and intrinsically resilient against desktop perturbations. With operational autonomy validated, **Phase-IV** addresses enterprise safety, regulatory policy, and cryptographic auditability.